DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND
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NAVAL SUPPLY SYSTEMS COMMAND
MECHANICSBURG PA 17055-0791

NAVSEAINST 4441.12A
NAVSUPINST 4441.37A
SEA 06L/NAVSUP N412
SEA 00B/340
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NAVSEA INSTRUCTION 4441.12A
NAVSUP INSTRUCTION 4441.37A

From: Executive Director, Naval Sea Systems Command
Vice Commander, Naval Supply Systems Command

Subj: MARITIME ALLOWANCE READINESS BASED SPARING, REPROVISIONING, AND CHANGE REVIEW BOARD

Ref: (a) OPNAVINST 4441.12D
(b) OPNAVINST 4442.5A
(c) NAVSUP P-488

Encl: (1) General Instructions Related to the MARRC-RB RBS System Checklist
(2) General Instructions Related to the MARRC-RB NON-RBS Major Reprovisioning Systems Checklist
(3) General Instructions Related to the MARRC-RB Allowance Change Request/Allowance Change Request-Fixed (ACR/ACR-F) Checklist
(4) General Instructions Related to the MARRC-RB Allowance Exception Request Checklist
(5) General Instructions Related To Using Interim Navy Common Readiness Model (NCRM) Tools
(6) Interim Navy Common Readiness Model (NCRM) Required Data Element Parameters
(7) Interim Navy Common Readiness Model (NCRM) Required Simulation Data Element Criteria
(8) Reliability Block Diagram Criteria

1. Purpose. To establish a joint Maritime Allowance Readiness Based Sparing (RBS), Reprovisioning, and Change-Review Board (MARRC-RB) to review and approve RBS optimizations and all major Naval Sea Systems Command (NAVSEA) reprovisioning efforts that result in changes to maritime allowances. This joint instruction establishes the allowance management oversight by Naval Supply Systems Command (NAVSUP) as required by Office of the Chief of Naval Operations (OPNAV) per references (a) and (b) as well as the associated funding and allowancing technical data by NAVSEA Logistics Director; Assistant Deputy Commander for Logistics Support (SEA 06L).

2. Cancellation. NAVSEAINST 4441.12 and NAVSUPINST 4441.37.

3. Scope and Applicability

a. This instruction applies to proposed maritime allowances resulting from:

(1) RBS optimizations, (initial, re-optimizations, and back-fits);

(2) Allowance Change Requests/Allowance Change Request-Fixed (ACR/ACR-F) as defined in reference (c);

(3) Allowance exception requests; and

(4) Major re-provisioning efforts. Major re-provisioning is defined as:

(a) An effort which will result in the increase range and depth of spare and repair part allowances of a platform (e.g., ship/shore activity Unit Identification Code), or of a major weapons system, by more than 30 percent; or

(b) An effort which will result in an Other Procurement Navy-8 (OPN-8), Weapons Procurement Navy (WPN-6) or Ship Construction Navy (SCN) buy-out of more than $250,000; or

(c) As directed by Director, Supply, Ordnance and Logistics Operations (OPNAV N41) or NAVSEA Program Executive Office (PEO).

b. Generally excluded are requirements determinations specifically governed by separate policy, such as TRIDENT Submarine (SSBN) Hull, Mechanical and Electrical (HME) and Strategic Weapons System (SWS) Coordinated Shipboard Allowance Lists (COSALs), nuclear reactor plant program material (Q COSAL) under the cognizance of NAVSEA Nuclear Propulsion office (SEA 08), bulk petroleum, and controlled cryptographic material.

4. Policy

a. The MARRC-RB will provide oversight and discipline of allowances arising from RBS, re-provisioning efforts, ACR/ACR-F, or allowance exception requests to ensure they are technically compliant, planned, budgeted/funded and valid per prevailing maritime allowance policy.

b. The MARCC-RB will identify potential policy issues that need to be reviewed and provide recommendations to the Allowance Working Group for consideration.

c. The MARRC-RB will be comprised of representatives from:

(1) NAVSUP Deputy Director, N41B for Supply Chain Management (Co-chair)

(2) NAVSEA 06L Deputy Director Readiness and Logistics (Co-chair)

(3) NAVSUP Weapon Systems Support (WSS) Maritime Allowance and Outfitting Department (N43)

(4) Naval Sea Logistics Center (NSLC) Life-Cycle Logistics Support Department Code 61

d. Resolution of specific issues may require expertise to address specific concerns and process changes. The MARRC-RB will engage and request participation as needed from:

(1) NAVSUP Business Systems Center (BSC)

(2) Commander, U. S. Fleet Forces Command (USFF) Fleet Supply (N41) and Commander, U. S. Pacific Fleet (COMPACFLT) Fleet Supply (N4)

(3) In-Service Engineering Agent (ISEA)

(4) Hardware Systems Command (HSC) /Program Offices

(5) Program Executive Offices (PEO)

(6) NAVSUP WSS Program Manager (PM)

(7) OPNAV

5. Responsibilities. The activities designated as primary participants in the MARRC-RB will assign a representative to serve as a member on the review board. The NSLC Gatekeeper (GK) will coordinate the scheduling and create the agenda for the meeting. The agenda will include the system RBS optimizations, ACR requests, allowance exceptions, and proposed major re-provisioning actions. Any member of the board or Program Offices, PEOs, and ISEAs may request an unscheduled/emergent meeting through the Review Board coordinator to address critical or urgent issues.

a. MARRC-RB will:

(1) Convene (monthly (second Tuesday of the month) or as required) meetings to review the pending RBS optimizations and major re-provisioning efforts.

(2) Per enclosures (1) through (8), deploy the checklists and procedures to facilitate the

approval or denial of RBS (initial, re-optimizations, and back-fits), re-provisioning efforts, ACR, and allowance exception requests (see paragraph 8 for access to updatable checklists).

(3) Communicate with stakeholders regarding MARRC-RB decisions.

(4) For RBS systems, validate the cost effectiveness of the presented Operational Availability (A_o). In this regard, the MARRC-RB may make recommendations to adjust the modeled A_o target during the system review.

(a) The Program Office, in conjunction with the OPNAV Warfare Systems Resource Sponsor (OPNAV N9), must be afforded the opportunity to meet and discuss the recommendations of the MARRC-RB to adjust the modeled A_o target and report their decision to the MARRC-RB.

(b) The HSC Program Office must coordinate the necessary action, with a system's OPNAV N9 Resource Sponsor, to obtain approval for a recommended deviation from any OPNAV A_o goal/value.

(5) Ensure all decisions and data files are provided to designated NAVSEA GK, NSLC, for processing.

(6) Any member can request an additional review if there are any issues with the Standard Allowance File Tool (SAFT) updates.

b. NAVSEA 06L will:

(1) Co-chair the MARRC-RB.

(2) Provide management and oversight for all actions associated with programming, coordinating, budgeting, justifying, defending, and executing requirements for OPN-8, WPN-6, and SCN outfitting spares requisitions.

(3) Ensure implementation of this policy and provide updates as needed.

c. NAVSUP will:

(1) Co-chair the MARRC-RB.

(2) Provide management and oversight for all actions associated with coordinating and executing Maritime allowance requirements.

(3) Provide oversight and guidance for NAVSUP WSS provisioning processes.

(4) Ensure implementation of this policy and provide updates as needed.

d. NAVSUP WSS N43 will:

(1) Refer all requests for allowance exception overrides to the NSLC GK for coordination and scheduling with the MARRC-RB.

(2) Monitor allowances to identify significant equipment/system specific allowance adds/increases and modifications to ensure validity.

(3) Report any suspected unauthorized/unapproved allowance Adds/Increases/Modifications to the NSLC GK for review by the MARRC-RB.

(4) Provide allowancing process technical oversight to ensure any decision made by MARRC-RB is functionally possible within the scope of the allowancing process.

(5) Develop and maintain a database repository for the upload and controlled access of all RBS, ACR/ACR-F, allowance exception, and re-provisioning requests including cross reference identification numbers, artifacts and details of the final MARRC-RB adjudication.

e. NSLC will:

(1) Serve as the NAVSEA point of entry ("Gatekeeper") for all requests for allowance changes.

(2) Serve as the coordinator for the MARRC-RB including scheduling and developing the agenda.

(3) Provide updates to the database repository for all submitted RBS, ACR requests, allowance exception requests, and re-provisioning requests, regardless of disposition, to include details of final MARRC-RB adjudication as well as the following:

(a) Coordinate with NAVSUP WSS to ensure an identification number is assigned by type and date to facilitate access to entries in the repository.

(b) The details of the request including copy of the checklist, MARRC-RB disposition and for RBS systems the following data:

1. Model

2. Reliability Block Diagram

3. Boundary of model

4. Mission profile

5. Configuration profile

6. RBS curve with target A_o

7. Part file

8. A_o reference (e.g. Capability Development Document/Capability Development Plan)

(4) Verify funding is available for allowance modifications, (adds/increases), as a result of RBS or re-provisioning.

(5) Process RBS, ACR/ACR-Fs and allowance modifications based on the MARRC-RB decisions, including forwarding Standard Allowance File (SAF) updates to NAVSUP WSS N43.

6. Records Management

a. Records created as a result of this instruction, regardless of format or media, must be maintained and dispositioned according to the records disposition schedules found on the Directives and Records Management Division (DRMD) portal page: https://portal.secnav.navy.mil/orgs/DUSNM/DONAA/DRM/SitePages/Home.aspx.

b. For questions concerning the management of records related to this instruction or the records disposition schedules, please contact your local Records Manager or the DRMD program office.

7. Review and Effective Date. Per OPNAVINST 5215.17A, NAVSEA 06L and NAVSUP N4 will jointly review this instruction annually around the anniversary of its issuance date to ensure applicability, currency, and consistency with Federal, Department of Defense, Secretary of the Navy, and Navy policy and statutory authority using OPNAV 5215/40 Review of Instruction. This instruction will be in effect for 10 years, unless revised or cancelled in the interim, and will be reissued by the 10-year anniversary date if it is still required, unless it meets one of the exceptions in OPNAVINST 5215.17A, paragraph 9. Otherwise, if the instruction is no longer required, it will be processed for cancellation as soon as the need for cancellation is known following the guidance in OPNAV Manual 5215.1 of May 2016.

8. Forms. The following forms are available for download from the Naval Forms Online

(NFOL) Web site at: https://navalforms.documentservices.dla.mil/web/public/home and the MARRC-RB Repository Web site at: https://my.navsup.navy.mil/apps/ops$rbs.home.

a. NAVSEA 4441/3 MARRC-RB RBS Systems Checklist

b. NAVSEA 4441/4 MARRC-RB Non-RBS Major Reprovisioning Systems Checklist

c. NAVSEA 4441/5 MARRC-RB ACR/ACR-F Checklist

d. NAVSEA 4441/6 MARRC-RB Allowance Exception Request Checklist

e. NAVSEA 4441/3 MARRC-RB Interim Navy Common Readiness Model (NCRM) Required OPUS Parameters

f. NAVSEA 4441/3 MARRC-RB Interim Navy Common Readiness Model (NCRM) OPUS10 Suite SIMLOX Data Element Criteria

g. NAVSEA 4441/3 MARRC-RB Reliability Block Diagram Criteria

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Releasability and Distribution:
This instruction is cleared for public release and is available electronically only via the Naval Logistics Library (NLL) Web site, https://nll.navsup.navy.mil/and via the NAVSEA Portal Web site, https://navsea.portal.navy.mil.

GENERAL INSTRUCTIONS RELATED TO THE MAARC-RB RBS SYSTEM CHECKLIST

1. The Maritime Allowance Readiness Based Sparing (RBS), Reprovisioning, and Change-Review Board (MARRC-RB) System Checklist will be utilized for initial and follow-on optimization actions for RBS systems.

a. The checklist form must be signed by the co-chairpersons and must indicate the approval or disapproval of the system's RBS analysis.

b. The checklist will be published and distributed with the minutes of the meeting and filed in the MARRC-RB repository.

2. The Hardware Systems Command/Program Executive Office (HSC/PEO) Program Managers (PM) must schedule a MARRC-RB review of their completed RBS analyses via the NSLC "Gatekeeper" (GK) at the earliest opportunity, but no later than 1 month prior to the desired update of the Standard Allowance File Tool (SAFT).

a. In addition to the required MARRC-RB member attendees, a member of the RBS team performing the actual RBS analysis should also attend as required.

b. Verify that the checklist was assigned a unique "Identification Tracking Number" and creation date for subsequent reference and access in the central MARRC-RB repository.

3. The HSC PM or designated member of the RBS analysis team will fill out and provide sections I through III of the checklist to the MARRC-RB no later than 2 weeks prior to the scheduled date of the review. If a contractor has performed the RBS analysis/optimization, specify contractor name and POC(s). Although most of the entries are self-explanatory, the following specifics apply to Section I:

a. RBS Types can be either Initial Optimizations or Re-optimizations. Backfits can apply to either of these types. If additional clarification is required beyond type, use the remarks block accordingly.

b. For Re-optimizations, please provide the most "Current Actual Operational Availability (A_o)" as well as the "Source" of this data and the "Reason" for the system's re-optimization.

c. For all RBS types, provide:

(1) "A_o Threshold" (aka, Minimum Acceptable A_o), the "A_o Target Objective" (e.g., the ultimate A_o value or target goal), and the Modeled A_o.

(2) Source and the document reference that authorizes/establishes the above A_0 values.

4. Upon MARRC-RB approval, the SAFT will be loaded with the resultant overrides and will include activation dates.

5. The MARRC-RB will validate the following:

a. System A_0 goal:

(1) Source of A_0 goal (e.g., Test and Evaluation Master Plan requirement, Capability Development Document /Capability Development Plan, Top Level Requirement, Weapon Spec, etc.)

(2) Accompanying documentation

(3) Reasonableness of A_0 goal

(4) Application of Multi-Tiered RBS

b. RBS model used:

(1) Navy approved (e.g., Interim Navy Common Readiness Model (NCRM) (OPUS Suite)), or,

(2) OPNAV approved alternative or OPNAV waiver letter for an unapproved model

c. Availability of Program Support Data.

d. NAVSUP WSS Buy-In funding and material availability.

e. NAVSEA 06L4 Buy-Out funding.

f. The A_0 value achieved by the system when zero spares are provided (this figure provides a baseline value for comparison with the target A_0 RBS analysis result).

g. The A_0 value achieved and associated cost savings by using Maintenance Assist Modules (MAMs) as spares.

h. The A_0 value achieved when RBS sparing (referred to as Enhanced RBS) is designated to augment/supplement the existing range/depth of authorized spares.

i. Multi-Tiered RBS A_0 (lesser of 5 percent below target A_0 or knee of curve) analysis including the cost savings (see reference (b), para 5.b and 7.a. (3) for further detail and definition).

j. Source and reasonableness of reliability parameters.

(1) Mean Time Between Failures (MTBF)

(2) Mean Time To Repair (MTTR)

(3) Mean Requisition Response Time

GENERAL INSTRUCTIONS RELATED TO THE
MARRC-RB NON-RBS MAJOR REPROVISIONING SYSTEMS CHECKLIST

1. The Maritime Allowance Readiness Based Sparing (RBS), Reprovisioning, and Change- Review Board (MARRC-RB) Non-RBS Reprovisioning Systems Checklist will be utilized for major reprovisioning efforts as defined in the MARRC-RB instruction.

2. The NAVSUP Weapon Systems Support (WSS) Program Manager (PM) in coordination with the Hardware Systems Command (HSC) Program Office and In-Service Engineering Agent (ISEA) will provide the initial notification of a proposed reprovisioning effort to the Naval Sea Logistics Center (NSLC) Gatekeeper (GK) providing sufficient data to fill out all of Section I and portions of Section II of the MARRC-RB Non-RBS Major Reprovisioning Systems Checklist.

a. To facilitate the assessment and determine the return on investment (ROI), break out of spares data by new installs and backfits for existing installs is required.

(1) For New Installs the full range, depth and cost of spares must be provided.

(2) For reprovisioning backfit/applied to existing installs, both the range/depth/cost of Adds and Increases as well as Deletes and Decreases (to existing sparing) must be provided.

3. Verify the checklist was assigned a unique Identification Tracking Number and creation date for subsequent reference and access in the central MARRC-RB repository.

4. The NAVSUP WSS PM must be prepared to present a brief summary of the reprovisioning effort to include justification and funding requirements.

5. The MARRC-RB will utilize this checklist form to document the proceedings and results of the reprovisioning validation review as follows:

a. The checklist must be signed by the co-chairpersons and must indicate the approval or disapproval of the system's reprovisioning results.

b. The checklist form will be retained by the MARRC-RB coordinator and will be distributed with the minutes of the meeting and filed in the MARRC-RB repository.

GENERAL INSTRUCTIONS RELATED TO THE MARRC-RB ALLOWANCE CHANGE REQUEST/ALLOWANCE CHANGE REQUEST-FIXED (ACR/ACR-F) CHECKLIST

1. The Maritime Allowance Readiness Based Sparing (RBS), Reprovisioning, and Change-Review Board (MARRC-RB) will review and approve Allowance Change Requests/Allowance Change Request-Fixed (ACR/ACR-F). The objective is to ensure recommended ACR/ACR-F approvals are supported by a thorough technical review and data (e.g., demand, maintenance, configuration, and criticality). Review criteria and thresholds for ACR/ACR-F to be determined by the review board.

2. The MARRC-RB must utilize this ACR/ACR-F checklist to document the proceedings and record the final decisions.

a. For ACRs, the Naval Sea Logistics Center Gatekeeper will provide the data to complete Section I of the checklist. Since ACRs are consumable and primarily Defense Logistics Agency low cost items, Section II of the checklist is not required.

b. For ACR-Fs, the NAVSUP Weapon Systems Support (WSS) Program Manager (PM) will provide sufficient data to complete Section I and, when the ACR-F is either an add or an increase, complete Section II.

Please note that the originally submitted ACR/ACR-F should include all the information that is required in Section I of this checklist and can be attached in lieu of filling out the checklist.

3. The MARRC-RB will ensure necessary funding and material are available to support all approved allowance changes generated by approved ACR/ACR-Fs.

4. The WSS PM will also load Planned Program Requirements for all approved ACR-F adds and increases as required.

GENERAL INSTRUCTIONS RELATED TO THE MARRC-RB ALLOWANCE EXCEPTION REQUEST CHECKLIST

1. The Maritime Allowance Readiness Based Sparing (RBS), Reprovisioning, and Change-Review Board (MARRC-RB) will review and approve/disapprove requests for overrides to manually adjust (Add/Increase or Delete/ Decrease) allowances originating from non-traditional sources (e.g., NAVSUP Weapon Systems Support (WSS) Program Managers (PMs), Hardware System Commands (HSCs) and In-Service Engineering Agents (ISEAs)). This excludes changes resulting from Fleet Allowance Change Requests/Allowance Change Request-Fixed (ACR/ACR- F), RBS optimizations and major re-provisioning efforts addressed in enclosures (1) through (3) of the MARRC-RB instruction.

2. At a minimum, the NAVSUP WSS PM (in coordination with the HSC and/or ISEA as necessary) will provide the initial notification of a proposed allowance exception/modification request to the Naval Sea Logistics Center (NSLC) Gatekeeper (GK), providing sufficient data to fill out all of Section I and Section II (for Adds/Increases only) of this checklist.

3. The NSLC GK will monitor and collect requests for manual overrides for MARRC-RB review and disposition. Additionally, the repository will automatically assign each checklist a unique Identification Tracking Number and creation date (verified by the NSLC GK) for subsequent reference and access in the central MARRC-RB repository.

4. Following completion of an upgraded and expanded NAVSUP Web based ACR program, which will require the submission of all allowance changes/exceptions as well as pertinent review data, the submitting activity can attach the Web based form for MARRC-RB review and approval. The latter eliminates the need to fill out Section I of this checklist.

5. For multiple line-item entries against a single system Allowance Parts List, submission of an attachment to the check list listing the items is acceptable. However, it should be noted that submissions of this nature may be considered a re-provisioning and follow the Non-RBS Major Reprovisioning Systems Checklist, enclosure (2), of the MARRC-RB instruction guidelines.

6. The NAVSUP WSS PM must be prepared to present a brief summary of the request to include justification and material/funding requirements (for adds/increases only).

7. The MARRC-RB must utilize the checklist form to document the proceedings and results of the review as follows:

a. The checklist must be signed by the co-chairpersons and must indicate the approval or disapproval of the requested allowance modification.

b. The MARRC-RB will also ensure necessary funding and material are available to support all approved allowance changes generated by approved allowance exception requests.

c. The checklist form will be retained by the MARRC-RB coordinator and will be distributed with the minutes of the meeting and filed in the MARRC-RB repository.

GENERAL INSTRUCTIONS RELATED TO USING
INTERIM NAVY COMMON READINESS MODEL (NCRM) TOOLS

1. These instructions provides additional guidance and data standards used within the Interim NCRM Suite to MARRC-RB Co-Chairs, Hardware Systems Command (HSC), Program Executive Office (HSC/PEO) Program Managers (PM), Program Acquisition Resource Managers (PARM), Technical Support Activities (TSA), In-Service Engineering Activities (ISEA) and other NCRM Practitioners for initial / updates to retail allowances, and for development of required authoritative Maritime Reliability Block Diagrams (RBD).

2. This guidance applies to proposed Maritime allowances submitted to the MARRC-RB review / approval process resulting from Interim NCRM Readiness Based Sparing (RBS) and SIMLOX derived allowance optimizations, such as initial, re-optimizations and back-fits.

3. As stated previously, generally excluded are requirements determinations specifically governed by separate policy, including TRIDENT Submarine (SSBN) Hull, Mechanical and Electrical (HM&E), Strategic Weapons System (SWS) Coordinated Shipboard Allowance Lists (COSALs), nuclear reactor plant program material (Q COSAL), bulk petroleum and controlled cryptographic material.

4. If the Interim NCRM is used for SSBN, HM&E, SWS COSAL, nuclear reactor plant program material, bulk petroleum and controlled cryptographic material, reference (a) and this MARRC-RB Instruction Supplement applies.

5. The OPNAV approved Maritime RBS model is transitioning to NCRM. The interim approved RBS model is the OPUS Suite. All Maritime system / equipment programs, acquisition category (ACAT) and non-ACAT, should use the NCRM interim OPUS Suite RBS model and associated Maritime required RBDs for all new, altered or updated systems / equipment sparing.

6. RBS Model Result Presentation. The system / equipment Product Support Manager (PSM), or designated representative, attendance at the MARRC-RB is strongly recommended for presenting RBS candidates submitted to the MARRC-RB. If questions arise, and the PSM or representative is not in attendance, RBS candidates will be deferred until the next MARRC-RB.

INTERIM NAVY COMMON READINESS MODEL (NCRM) REQUIRED DATA ELEMENT PARAMETERS

Required OPUS Parameters			
Table	**Column ID**	**Description**	**Input**
ControlParameters	STYPE	Scenario type	STEADY-STATE
	PTYPE	Spares problem type	INITIAL
	PRMOE	Primary MoE	A
	NPOINT	Number of points	50
	MINMOE	Minimal MoE	0
	MAXMOE	Maximal MoE	1
GlobalParameters	SCLEN	Scenario length	0
	SCTIM	Scenario Start Time	0
	CURRN	Currency name	$
System	SID	System Identifier	Insert System Name
Item	IID	Item identifier	NIIN*
	DESCR	Description	EQUIPMENT NOMEN*
	PRICE	Unit price	UNIT PRICE*
	FRT	Failure rate	"=(1000000*BRF)/8760"*
	TYPE	Type	DU
ItemStructure	IID	Subitem identifier	NIIN*
	MID	Motheritem or system identifier	Insert System Name
	QTYPM	Quantity per motheritem	Integer Number, varies for each item*
Station	STID	Station identifier	"WHOLESALE"; "SHIP STOREROOM"; "SHIP"
	QTY	Total quantity	1
	TYPE	Type	DEPOT;OP
	MAXST	Maximal Stock	0 for "Wholesale" and "Ship"
StationStructure	STID	Station identifier	"SHIP";"SHIP STOREROOM"
	MSTID	Mother station identifier	"SHIP STOREROOM";"WHOLESALE"
	TFRMS	Time from mother station	2.0; 360
	TTOMS	Time to mother station	2.0; 360
SystemDeployment	SID	System identifier	Insert System Name
	USTID	Unit or station identifier	"SHIP"
	QTYPS	Quantity per unit or station	1
	UTIL	Utilization per calendar time	8760
SystemRepair	SID	System Identifier	Insert System Name
	STID	Station Identifier	"SHIP"
	SUROT	Subitem Replacement TAT (MTTR)	2
	SURPF	Subitem Replacement Item Fraction	1
ItemReorder	IID	Item identifier	NIIN*
	STID	Station identifier	"WHOLESALE"
	LEADT	Lead time	0
	MINRS	Minimal reorder size	MRU*

Optional OPUS Parameters			
Table	Column ID	Description	Input
Redundancy	RLID	Redundancy Link Identifier	Insert unique Identifier for Redundancy Link
	SID	System Identifier	Insert System Name
	NRL	Number of Redundancy Links	Interger number of links
	NRLRQ	Number of Required links	Integer number of links required
	ACPA	Active/Passive Option	ACTIVE; PASSIVE
	TSWOV	Time to switch over	Delay time to switch to redundant link
RedundancyLink	RLID	Redundancy Link Identifier	Insert unique Identifier for Redundancy Link
	IID	Item identifier	NIIN*
	QTYPL	Quantity per link	Quanitity of each item in a single redundancy link

* **Note:** For values of **ITEM FRT** (Failure Rate) and **System Deployment UTIL** (Utilization per Calendar Time) use actual system / equipment operational / actual run-time; if the operational / actual run-time is not accurately known use the default value of 8,760.

INTERIM NAVY COMMON READINESS MODEL (NCRM) REQUIRED SIMULATION DATA ELEMENT CRITERIA

Required SIMLOX Parameters			
TABLE	**Column ID**	**Opus Suite Description**	**Input**
Control	NREPS	Number of replications	100
	SIMPE	Simulation period	From DRMP
	RSEED	Random seed	123456789
	APID	Allocation Point Identifier	Stock Allocation Point Identifier (from OPUS C/E curve)
	RCINT	Result collection interval	0.5
	RCSTA	Result collection start time	0
	RELOP	Reliability option	RBD; REDUNDANCY
MissionType	MTID	Mission type identifier	Mission Phase Identifier
	NOS	Nominal number of systems	Quantity of Systems being analyzed
	MNOS	Minimum number of systems	Quantity of Systems being analyzed
	DURN	Mission duration	From DRMP
	MPAT	Mission Pattern	CONTINUOUS
OperationProfile	PRID	Profile identifier	Mission Profile Identifier
	SPRID	Subprofile identifier	MTID from MissionType
	STIM	Start time	From DRMP
	ITYPE	Initiation type	SCHEDULED
	IQTY	Initiation quantity	1
Operations	USTID	Operational Mode Identifier	"SHIP"
	PRID	Profile identifier	Mission Profile Identifier
StockAllocation	POINT	Point Identifier	Stock Allocation Point Identifier (from OPUS C/E curve)
	IID	Item identifier	NIIN
	STID	Station identifier	"SHIP STOREROOM"
	STSIZ	Stock size	Quanitity of items allowed
	ROSIZ	Reorder size	MRU

Optional SIMLOX Parameters			
TABLE	**Column ID**	**Opus Suite Description**	**Input**
RBDParallel	NBREQ	Number of blocks required	
	NBOP	Number of blocks in operation	
	TSWOV	Time to switch over	
	ONLM	Online maintained	
	RBID	Parallel block identifier	
RBDParallelStructure	SRBID	Sub block identifier	
	OPMID	Operational mode identifier	<DEFAULT>
	QTYPB	Quantity per block	
	RBID	Series block identifier	
RBDSeries	RBID	Series block identifier	
RBDSeriesStructure	SRBID	Sub block identifier	
	OPMID	Operational mode identifier	<DEFAULT>
	QTYPB	Quantity per block	
	POINT	Point identifier	
OperationalModes	OPMID	Operational Mode Identifier	
MissionOperationalModes	MTID	Mission Type Identifier	
	SID	System Identifier	
	OPMID	Operational Mode Identifier	
MaterialOperationalModes	MID	Materiel Identifier	
	OPMID	Operational mode identifier	<DEFAULT>
	RBID	Reliability block identifier	
	USERT	Usage Rate	<1.0>

RELIABILITY BLOCK DIAGRAM (RBD) CRITERIA

1. This enclosure describes criteria for developing Maritime standardized ship system RBDs.

a. RBDs are used primarily for input to Readiness Based Sparing (RBS) related simulation and analytic models to calculate system reliability and operational availability (A_o). These RBDs within RBS models are typically used to determine maintenance repair parts sparing to meet those reliability and A_o metrics at affordable cost or as an analysis metric showing a platform's operational availability over select time periods.

b. The RBD will be developed/designed in early acquisition and updated to 'As Delivered' configuration after Post Shakedown Availability/Delivery. Similar to Configuration records, engineering changes and alterations that change the equipment/system data update/revise the RBD. A record of the RBD will be saved along with justification for the RBD update for historical purposes.

c. RBD depicts system elements and system operation and denotes the operational modes and redundancy of the elements for the platform modeled. Systems / equipment modeled for each platform can include propulsion, electrical, auxiliary, steering, navigation, combat, exterior and interior communications, etc.

d. RDBs are typically a pictorial form of a system showing interdependencies among mission essential system elements (subsystem, equipment, etc.). A complete understanding of the system's mission is required to produce a valid RBD. In addition to the pictorial representation of a system, reliability engineers should provide the RBD in table format to assist practitioners in establishing data for simulations.

2. General Requirements

a. RBDs are linked to Design Reference Mission (DRM) which defines the percentage of time during a mission phase that the equipment / system is used during execution of the function performed. For example, a surface search radar may only use short pulse transmission to improve close target recognition during sea and anchor detail which is a smaller subset (percentage) of operational time than transit. The RBD would take into account the operating requirements for different mission phases. Each system should have a top-level RBD representing its configuration for each phases of the mission profile for the platform.

b. Across the Maritime Enterprise only ONE RBD is used for an equipment / system to platform / unit configuration. For example the RBD for an identical AN/SQQ-89 on a DDG51 Flight II Class destroyer will have a different RBD if the AN/SQQ-89 is installed on a DDG51 Flight I Class Destroyer. Further, if warranted, configurations may vary within a specific class of platform such that the RBD for an identical AN/SQQ-89 installation on the DDG51 would require another RBD for the same AN/SQQ-89 installed on the DDG52.

c. Care must be given by those developing an RBD to ensure that operational success paths among all elements of the system are correctly described and proper boundaries identified for each instance of installation, i.e., different RBD per equipment / systems and platform if warranted.

d. Care must also be taken to ensure the RBDs for similar equipment / systems have identical boundaries to ensure consistent A_0 metrics.

3. An RBD will include the following:

a. RBD number and descriptive title

b. Logic diagram or diagrams (blocks)

(1) Breakout of physically distinct equipment with identifiable Expanded Ship Work Breakdown Structure (ESWBS) or Functional Group Code.

(2) Engineering reliability and maintainability data, i.e., Mean Time Between Failure (MTBF), Mean Time to Repair (MTTR), Duty Cycle, Total Available Downtime, etc.

(a) Note: Generation of readiness indices at the system level is complex and requires calculation of downtime and associated downtime indices with special attention to overlapping events and not simple counts of failure and downtime

(3) Correctly describe equipment / system functionality as series, parallel or in some cases extended parallel.

(4) Block assumptions include maintenance dependencies such as:

(a) Redundancy only

(b) Repair independent of non-failed operating blocks, etc.

(5) Block variables include:

(a) Time

(b) Cycles

(c) Events, etc. Phases matrices

1. DRM criteria typically is a day-by-day time-phased description of the operations, engagements and mission areas that the platform is expected to undergo while deployed for 120-days.

2. DRMs are generated for ship class and approved by CNO

a. Recently platforms are re-directed and perform missions that exceed or change DRMs, which in turn may impact RBD validity.

3. Equipment Reference Profile is a series of duty cycles for each RBD Block by mission phases.

a. Multiple sets of duty cycles may be required to describe multi-mission systems.

(6) Reliability blocks typically utilizes three main formulas;

(a) Reliability;

1. Reliability(Mission Time) = e-λt

2. λ (lambda) = Failure rate = 1/MTBF

3. t = mission time in cycles, hours, miles, etc.

4. e = natural logarithm = 2.71828

(b) Reliability(System) = R1 x R2 x R3 x R4 x ....RN

c. Reliability(Active Redundant Parallel System) = 1 – (1 – R1)(1 – R2)RBDs using mission free profiles (DRM not considered) are not generally valid for showing trends in material readiness,

(1) NAVSEA Policy requires availability calculations based on DRM and an equipment reference profile.

d. All RBDs have approval and a prepared by activity and date associated with the official data.

e. Minimum RBD artifacts include (but not limited to):

(1) Time meter assignments / runtime assumptions

(2) The parameters used to calculate the duty factors (Df / K factors)

(3) The DRM profile / DRM

(4) Parts level

(a) Parts to block assignment -> outlining the critical and non-critical parts / Line-Replaceable Unit

(b) Part Cost (Turn in Cost or new unit replacement)

(c) Failure Mode Effects and Criticality Analysis / Level of Repair Analysis

(d) Depot / Intermediate level repair / Organizational level repair

(e) Failure rates

1. Engineering Estimates or Actuals or both

(f) Mean Supply Response Time / Mean Down Time, MTTR

1. NAVSUP policies or Actuals or both

(g) System boundaries based on (Configuration Data Managers Database-Record Index Number, ESWBS, Allowance Parts List, End Item Code)

(h) Interoperability designs showing nodes and connections between systems.